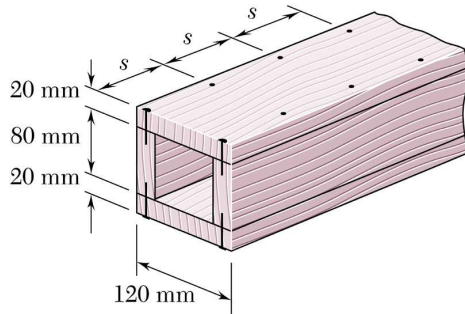


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6.3 A square box beam is made of two 20 x 80 mm planks and two 20 x 120 mm planks nailed together as shown. Knowing that the spacing between the nails is $s = 50$ mm, and that the allowable shearing force in each nail is 300 N, determine (a) the largest allowable vertical shear in the beam, (b) the corresponding maximum shearing stress in the beam.



Given:

- $s = 50$ mm
- $V_{per\ nail} = 300$ N
- $n = \#$ of nails per length $s = 2$

Find:

- Max allowable vertical shear
- Max shearing stress in the beam

Approach:

- I will use the equation for shear flow to find the max allowable vertical shear load by equating it with the shear load per unit length due to the nails

$$q_{all} = \frac{nV_{per\ nail}}{s} \left[\frac{N}{mm} \right]$$

- Equating this with the shear flow due to a vertical shear load V

$$q = \frac{VQ_{top}}{I}$$

- The max shearing stress (within the beam) is at the neutral axis and is given by

$$\tau_{max} = \frac{VQ}{It}$$

1

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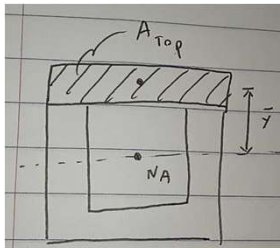
(a) Finding the moment of inertia by superposition

$$I_{outside} = \frac{1}{12} b_o h_o^3 = \frac{1}{12} [120\ mm][120\ mm]^3 = 17280000\ mm^4 = 0.00001728\ m^4$$

$$I_{inside} = \frac{1}{12} b_i h_i^3 = \frac{1}{12} [80\ mm][80\ mm]^3 = 3413333.33\ mm^4 = 0.000003413\ m^4$$

$$I = I_{outside} - I_{inside} = 1.386 \times 10^{-5} m^4$$

Finding the first moment of area of the top plank ($Q_{top} = Q_{bot}$)



$$\bar{y} = [40\ mm] + \frac{1}{2} [20\ mm] = 50\ mm$$

$$A_{top} = [20\ mm] \cdot [120\ mm] = 2400\ mm^2$$

$$Q_{top} = \bar{y} A_{top} = 1.2 \times 10^{-4} m^3$$

Finding the allowable shear flow due to nails

$$q_{all} = \frac{nV_{per\ nail}}{s} = \frac{2[300\ N]}{[0.05\ m]} = 12000\ \frac{N}{m}$$

2

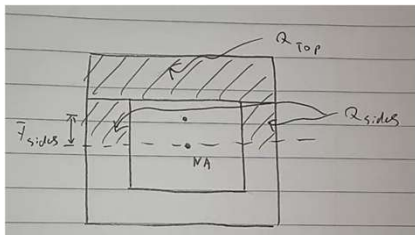
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Find max allowed vertical shear

$$q = \frac{VQ_{top}}{I}$$

$$V_{max} = \frac{q_{all} I}{Q_{top}} = \frac{\left[12000 \frac{N}{m}\right] [1.386 \times 10^{-5} m^4]}{[1.2 \times 10^{-4} m^3]} = 1386 N$$

(b) Finding the first moment of area of the top half using superposition



$$Q_{sides} = \bar{y}_{sides} A_{sides} = [20 mm] \cdot 2[40 \cdot 20 mm^2] = 3.2 \times 10^{-5} m^3$$

$$Q = Q_{top} + Q_{sides}$$

$$Q = [1.2 \times 10^{-4} m^3] + [3.2 \times 10^{-5} m^3] = 1.52 \times 10^{-4} m^3$$

This result makes sense since an increase in the first area moment of the top plank (Q_{top}) will increase the shear flow (q) because there is more material trying to shear the nails. Thus, this would cause the max allowable shear to decrease: $V_{max} = \frac{q_{all} I}{Q_{top}}$

$$[Time spent: 45 mins]$$

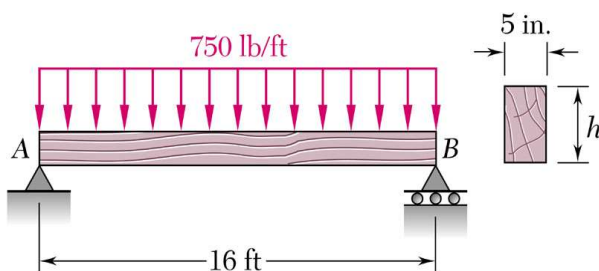
Find max shear stress

$$\tau_{max} = \frac{VQ}{It} = \frac{[1386 N][1.52 \times 10^{-4} m^3]}{[1.386 \times 10^{-5} m^4][2 \cdot 0.02 m]} = 380000 \frac{N}{m^2}$$

3

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6.15 For the beam and loading shown, (a) sketch profiles of normal and shear stresses on a beam cross-section, and (b) determine the minimum required depth h , knowing that for the grade of timber used, $s_{all} = 1750$ psi in the direction of the wood grain, and $t_{all} = 130$ psi for shear stress parallel to the wood grain.



Given:

- $w = 750 \text{ lb/ft}$, $L = 16 \text{ ft}$
- $\sigma_{all} = 1750 \text{ psi}$
- $\tau_{all} = 130 \text{ psi}$

Find:

- Normal and shear stress profiles on the cross section
- Minimum depth h

Approach:

- The normal and shear stress profiles are described by the following equations:

$$\sigma = -\frac{My}{I}, \quad \tau = \frac{VQ}{It}$$

- To determine the minimum h , we first find the maximum bending moment and shear. Then we use the allowable section modulus and shear stress to find h depending on the limiting cases (normal or shear stress):

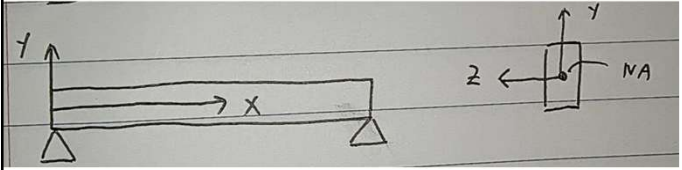
$$S = \frac{I}{c} = \frac{M_{max}}{\sigma_{all}}$$

$$\tau_{all} = \frac{V_{max} Q}{Ib}$$

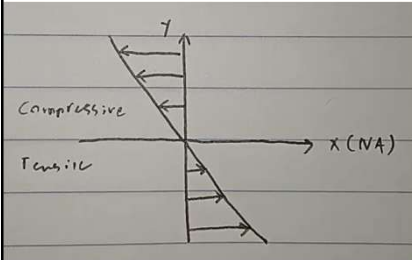
4

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(a) At a beam cross section, the internal bending moment and shear is constant (M, V)



Normal stress profile:



$$\sigma = -\frac{My}{I}$$

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Shear stress profile:

$$\tau = \frac{VQ}{It}$$

$$I = \frac{1}{12}bh^3 = \frac{1}{12}b(2c)^3 = \frac{2}{3}bc^3$$

$$t = b$$

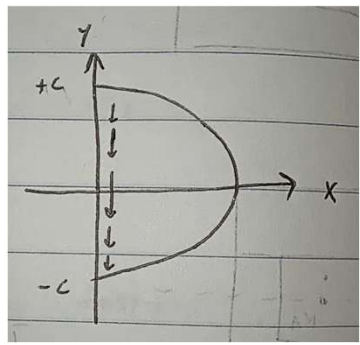
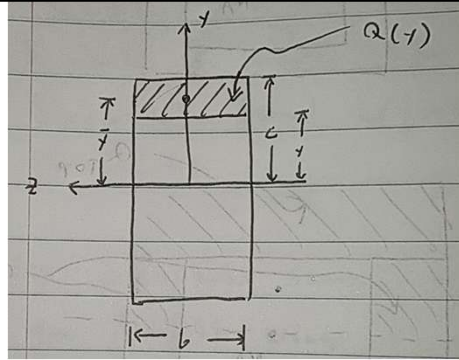
$$Q(y) = \bar{y}A(y)$$

$$Q(y) = \left(y + \frac{c-y}{2}\right)[(c-y)b] = \left(\frac{c+y}{2}\right)(c-y)b = \frac{b}{2}(c+y)(c-y)$$

$$Q(y) = \frac{b}{2}(c^2 - y^2)$$

$$\rightarrow \tau = \frac{VQ}{It} = V \frac{\frac{b}{2}(c^2 - y^2)}{\frac{2}{3}bc^3 \cdot b} = V \frac{3}{4bc^3}(c^2 - y^2)$$

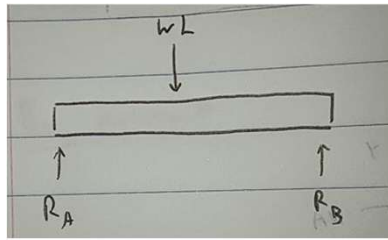
Note: V, b, c are constants. Shear stress is max at $y=0$ (neutral axis)



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(b) Find the reaction forces:



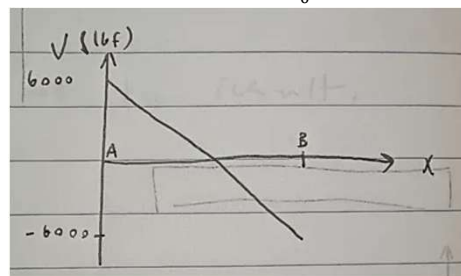
$$\sum M_A = 0 = R_B L - wL \left(\frac{L}{2} \right) \rightarrow R_B = 6000 \text{ lbf}$$

$$\sum F_y = 0 = R_A + R_B - wL \rightarrow R_A = 6000 \text{ lbf}$$

Thus, the magnitude of shear at A and B must equal R_A and R_B

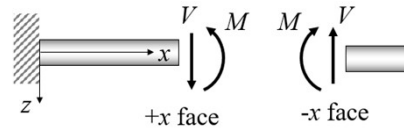
Starting from A, using the differential form:

$$\frac{dV}{dx} = -w \rightarrow V = V(0) + \int_0^x -w dx'$$



$$V_{max} = 6000 \text{ lbf}$$

Note: Although R_B (and thus V at B) is pointing upwards, by the sign convention on the $+x$ face, upward is negative V . (At A, we are looking at the $-x$ face)



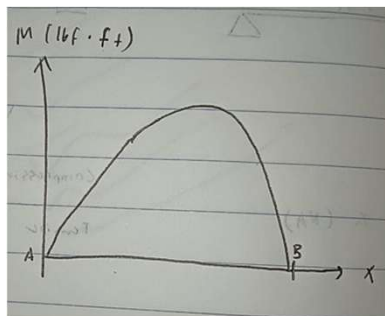
7

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(b) Starting from A, using the differential form:

$$\frac{dM}{dx} = V \rightarrow M = M(0) + \int_0^x V dx', V = -wx + 6000$$

$$M = 6000x - \frac{1}{2}wx^2$$



Note: $M(0)=0$ since simple supports don't contribute moment reactions. Also, M is max when $V=0$. Thus

$$M_{max} = M(x = 8) = 24000 \text{ lbf} \cdot \text{ft}$$

$$M_{max} = 288000 \text{ lbf} \cdot \text{in}$$

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If normal stress is the limiting case:

$$S = \frac{I}{c} = \frac{\frac{1}{12}bh^3}{\frac{h}{2}} = \frac{1}{6}bh^2$$

$$S = \frac{1}{6}bh^2 = \frac{M_{max}}{\sigma_{all}}$$

$$h = \sqrt{\frac{6M_{max}}{b\sigma_{all}}} = \sqrt{\frac{6[288000 \text{ lbf} \cdot \text{in}]}{[5 \text{ in}][1750 \text{ lbf/in}^2]}} = 14.052 \text{ in}$$

Picking the larger h:

$$\boxed{h = 14.052 \text{ in}}$$

Since the values of the depth is relatively close for both limiting cases, the beam is designed efficiently. If we picked the smaller h, then the beam would've failed under bending at the middle, where the bending moment is max. [Time spent: 45 mins]

If shear stress is the limiting case (Q for upper half from NA):

$$Q(y = 0) = \frac{b}{2}(c^2)$$

$$\tau_{all} = \frac{V_{max}Q}{Ib} = \frac{V_{max}\left(\frac{1}{2}bc^2\right)}{\left(\frac{1}{12}bh^3\right)b} = 6\frac{V_{max}\left(\frac{h}{2}\right)^2}{bh^3} = \frac{3}{2}\frac{V_{max}}{bh}$$

$$h = \frac{3V_{max}}{2\tau_{all}b} = \frac{3[6000 \text{ lbf}]}{2[130 \text{ lbf/in}^2][5 \text{ in}]} = 13.846 \text{ in}$$

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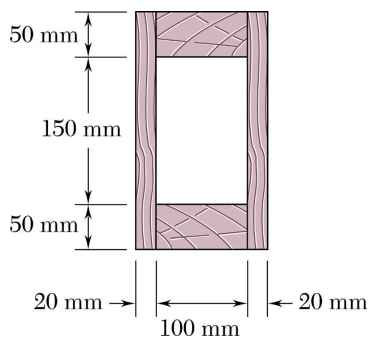
(GRADED) 6.31 The built-up box beam shown at left is made by gluing together two 20 x 250 mm plywood strips and two 50 x 100 mm planks. Knowing that the allowable average shearing stress in the glued joints is 350 kPa, determine the largest permissible vertical shear in the beam.

Given:

- $\tau_{ave,all} = 350 \text{ kPa}$

Find:

- Max allowable vertical shear in the beam



Approach:

- The max allowable vertical shear in the beam is limited by the allowable shearing stress in the glued joints. We relate these using the formula :

$$\tau = \frac{VQ}{It}$$

- Since we are looking at the stress at the joints, Q corresponds to the 50x100 mm plank. This is the piece that is being held up by the joint.
- Likewise, t is the total height of the contact surfaces where the glue resists the sliding.

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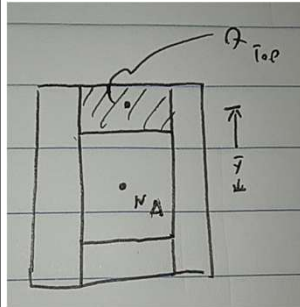
Finding the moment of inertia by superposition

$$I_{outside} = \frac{1}{12} b_o h_o^3 = \frac{1}{12} [140 \text{ mm}][250 \text{ mm}]^3 = 0.0001822 \text{ m}^4$$

$$I_{inside} = \frac{1}{12} b_i h_i^3 = \frac{1}{12} [100 \text{ mm}][150 \text{ mm}]^3 = 0.00002812 \text{ m}^4$$

$$I = I_{outside} - I_{inside} = 1.541 \times 10^{-4} \text{ m}^4$$

Finding the first moment of area of the top plank



$$\bar{y} = [75 \text{ mm}] + \frac{1}{2} [50 \text{ mm}] = 100 \text{ mm}$$

$$A_{top} = [50 \text{ mm}] \cdot [100 \text{ mm}] = 5000 \text{ mm}^2$$

$$Q_{top} = \bar{y} A_{top} = 5 \times 10^{-4} \text{ m}^3$$

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Find max allowable vertical shear

$$\tau_{ave,all} = \frac{VQ}{It} \rightarrow V = \frac{\tau_{ave,all} It}{Q}$$

$$V = \frac{[350 \times 10^3 \text{ N/m}^2][1.541 \times 10^{-4} \text{ m}^4][2 \cdot 0.05 \text{ m}]}{[5 \times 10^{-4} \text{ m}^3]}$$

$$V = 10787 \text{ N}$$

If the upper plank had a smaller area, Q would decrease and cause the max allowable vertical shear to increase. This makes sense since layers of material within a smaller plank would have less tendency slide relative to each other and thus result in less shear stress. If the contact area of the glue (represented by t) increases, then the max allowable shear should increase as well.

In terms of loading, the two 20 mm x 250 mm strips carry most of the vertical shear force since the shear stress is highest at the neutral axis. Thus, the shear loads are transmitted from these strips to the 50 mm x 100 mm planks via the glued joints and the shear flow must be continuous across this joint.

[Time spent: 35 mins]

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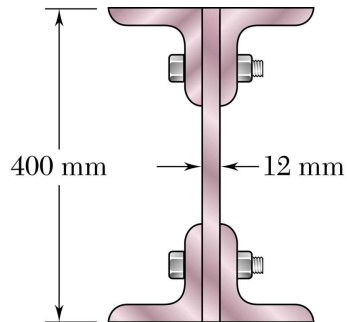
6.46 Four 102 x 102 x 9.5 steel angle shapes and a 12 x 400 mm steel plate are bolted together to form a beam with the cross-section shown on the right. The bolts are of 22-mm diameter and are spaced longitudinally every 120 mm. Knowing that the beam is subjected to a vertical shear of 240 kN, determine the average shearing stress in each bolt.

Given:

- $d = 22 \text{ mm}$
- $s = 120 \text{ mm}$
- $V = 240 \text{ kN}$

Find:

- Average shearing stress in each bolt



Approach:

- Using continuity of shear flow from the angle shapes to the steel plate, I can relate the shear flow from the angle to shear flow from the bolt:

$$q_{\text{angle}} = \frac{VQ_{\text{angle}}}{I}$$

$$q_{\text{bolt}} = \frac{nF_{\text{bolt}}}{s}$$

Where n is the number of bolts per length s

- Since the shear flow from 2 angle shapes combine to become the shear flow through the bolt, I can then find the force on the bolt:

$$2q_{\text{angle}} = q_{\text{bolt}}$$

- Finally, I can find the average shear stress in each bolt, noting that the bolt is under *double* shear.

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For a 102 x 102 x 9.5 steel angle shape (Appendix C):

- $A = 1850 \text{ mm}^2$
- $I = 1.80 \times 10^6 \text{ mm}^4$
- $y = 28.7 \text{ mm}$
- $\bar{y} = [200 - 28.7 \text{ mm}] = 171.3 \text{ mm}$

First moment of area of the angle (Q_{angle}):

$$Q_{\text{angle}} = \bar{y}A = 3.169 \times 10^{-4} \text{ m}^3$$

Moment of inertia:

Parallel axis theorem for each angle:

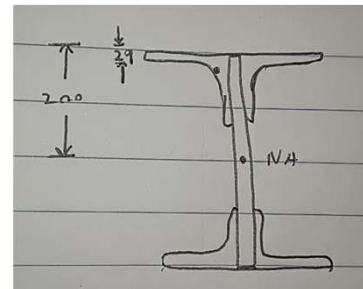
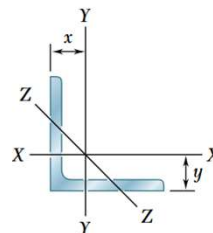
$$I_{\text{angle}} = I + A\bar{y}^2 = [1.80 \times 10^6 \text{ mm}^4] + [1850 \text{ mm}^2][171.3 \text{ mm}]^2 = 56.0858 \times 10^6 \text{ mm}^4$$

Superposition:

$$I_{\text{tot}} = \frac{1}{12}bh^3 + 4I_{\text{angle}} = \frac{1}{12}[12 \text{ mm}][400 \text{ mm}]^3 + 4[56.0858 \times 10^6 \text{ mm}^4] = 2.8834 \times 10^{-4} \text{ m}^4$$

Shear flow of angle:

$$q_{\text{angle}} = \frac{VQ_{\text{angle}}}{I} = \frac{[240 \times 10^3 \text{ N}][3.169 \times 10^{-4} \text{ m}^3]}{[2.8834 \times 10^{-4} \text{ m}^4]} = 263771.935 \text{ N/m}$$



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Shear flow of bolt:

$$q_{bolt} = \frac{nF_{bolt}}{s} = \frac{F_{bolt}}{[0.12 \text{ m}]}$$

$$2q_{angle} = q_{bolt} = \frac{F_{bolt}}{[0.12 \text{ m}]}$$

$$F_{bolt} = 63305.264 \text{ N}$$

Avg shear stress on bolt under double shear:

$$\tau_{ave} = \frac{F_{bolt}}{2A_{bolt}} = \frac{F_{bolt}}{2\left(\frac{\pi}{4}d^2\right)} = \frac{2}{\pi} \frac{[63305.264 \text{ N}]}{[0.022 \text{ m}]^2}$$

$$\tau_{ave} = 83267320.56 \text{ N/m}^2$$

$$\tau_{ave} = 83.267 \times 10^6 \text{ N/m}^2$$

In this design, there are 2 critical regions on the bolt since it is under double shear. It also makes sense that with an increase in the shear flow from the angle, the force on the bolt will have to increase to carry it. [Time spent: 35 mins]